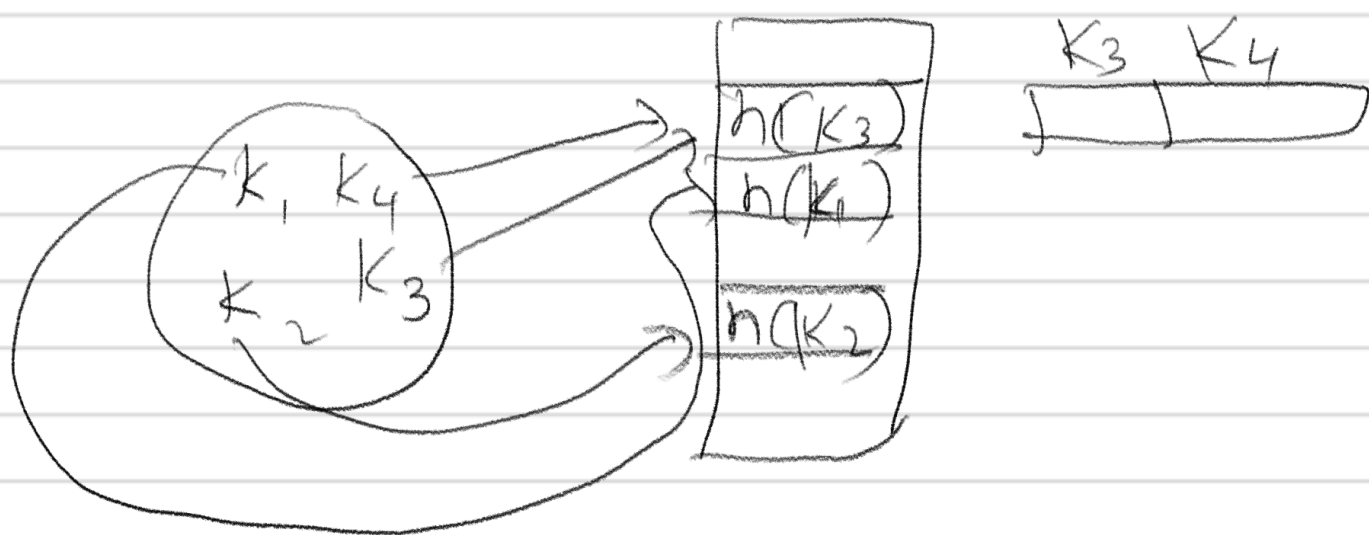


1) Direct addressing

- a) as many entries in the array as keys, lots of empty spaces
- b) Array index obtained from the key
- c) with a good hash function
- o (i) runtime
- d) chaining deals with collisions, open addressing deals with collisions



Good hash function

- Minimizes collisions
- A key hashes to any of the available slots independent of where any other k has hashed to

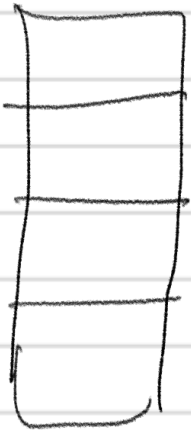
Open addressing

- 1) Every element occupies one slot
- 2) Table can become full
- 3) Frequency and number of keys - good to know
- 4) No linked list, better caching

Example hash function

- Key mod 5

50, 70, 76, 85



$$50 \% \text{ mod } 5 = 0$$

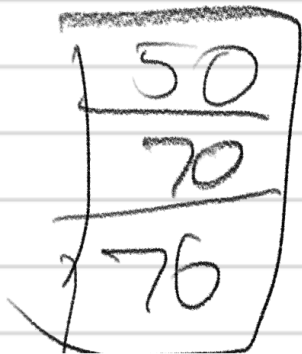
$$1) \text{ slot } 0 = 50$$



$$2) 70 \text{ mod } 5 = 0$$

but slot 0
is occupied

3) $76 \text{ mod } 5 = 1$ so try
slot 1 is
occupied from next
empty slot



4) 85 $85 \text{ mod } 0$
is 0, goes to
slot 3

Linear probing

hash(key)

if slot is occupied

hash(key) + 1

if occupied

hash(key) + 1 + 1

Quadratic probing

hash(key)

if slot is occupied

hash(key) + (next slot index)²

if slot is occupied

hash(key) + (next slot index
+ 1)²